복사전달 특강
Special Topics in Radiative Transfer

Lecture 10
2026 May 12 (Tuesday), 1PM

updated on 05/11

Based on: Chapter 8 of Noebauer & Sim (Living Reviews in Computational Astrophysics)

선광일 (Kwangil Seon)
KASI / UST

Chapter 8: MCRT in Outflows and Explosions

Moving media — reference frames, Sobolev approximation, expansion work

- Previous chapters assumed static media or low-velocity environments

Many astrophysical systems involve **large material velocities**: stellar winds, SN ejecta, accretion disk winds

Special relativistic effects — especially the **Doppler effect** — become crucial

- Key challenge: frequency-dependent opacities must be evaluated in the correct frame

8.1 Mixed-Frame Approach

- Two frames: Lab Frame (LF) and Comoving Frame (CMF)
- Packet propagation in LF
- Interactions computed in CMF
- Doppler + aberration transforms

8.2 Line Interactions

- Sobolev approximation
- Optical depth integration simplified
- Re-emission direction sampling
- Test: H Lyman α profile

8.3 Expansion Work

- Energy exchange in moving media
- Packets gain/lose energy in LF
- MCRT naturally captures expansion work

Why Moving Media Require Special Treatment

Breakdown of the static-medium approximation

In **static media**: opacity χ is frequency-independent over the packet trajectory → simple integration

In **moving media**: CMF frequency ν_0 changes continuously as the packet propagates through a velocity gradient

• The optical depth integral (Eq. 79) must account for the Doppler-varying opacity along the path

$\tau(l) = \int_0^l d\nu_0/dl \cdot (\nu/\nu_0) \cdot \chi_0(\nu_0)\, dl$ [Eq. 79, Noebauer & Sim]

The factor ν/ν_0 transforms the opacity from CMF to LF; $d\nu_0/dl$ encodes the Doppler shift along the path.

General Case: Computationally Expensive

- Must integrate frequency change along trajectory
- Opacity varies strongly near line resonances
- Requires re-evaluation at each propagation step

Sobolev Approximation (Sect. 8.2)

- Velocity field monotonically varying → each line resonance encountered once
- Reduces to a purely local problem at the Sobolev point
- Enables dramatic computational speedup

8.1 The Mixed-Frame Approach

Lab Frame (LF) propagation + Comoving Frame (CMF) interactions

Lab Frame (LF): defined so the computational grid is at rest — natural for tracking positions and times

Comoving Frame (CMF): local rest frame of the material — interaction physics takes simplest form

- Mixed-frame: propagate packets in LF, but transform to CMF for each interaction

Key idea: the handling of different tasks in MCRT simulations is easier in one or the other frame.
→ Spatial/temporal grid: defined in LF | Opacity/emissivity: evaluated in CMF

Doppler Shift (LF → CMF)

$\nu_{0,i} = \gamma \nu_i (1 - \beta \cdot n_i)$ [Eq. 71]

- $\beta = v/c$ (material velocity)
- $\gamma = (1 - \beta^2)^{-1/2}$
- $\nu_{0,i}$: incident CMF frequency
- n_i: incident propagation direction (LF)

Energy Transform (LF → CMF)

$\varepsilon_{0,i} = \gamma \varepsilon_i (1 - \beta \cdot n_i)$ [Eq. 72]

- Coherent scattering in CMF: $\varepsilon_{0,e} = \varepsilon_{0,i}$
- Re-transform after interaction (Eqs. 75–76):

$\varepsilon_e = \gamma \varepsilon_{0,e} (1 + \beta \cdot n_{0,e})$

$\nu_e = \gamma \nu_{0,e} (1 + \beta \cdot n_{0,e})$

8.1 Aberration and Packet Initialisation

Direction transforms and CMF initialisation of packets

Aberration: propagation direction transforms between LF and CMF (in addition to Doppler)

- Must be taken into account when converting incident direction to CMF and emergent direction back to LF

Direction Transform LF → CMF [Eq. 77]

$n_{0,i} = (\nu/\nu_0) \cdot [\, n_i - (1 - c\,\gamma / (c\,\gamma + 1)\, n_i \cdot v)\, \gamma\beta \,]$

- Used to convert direction before interaction
- Necessary for O(v/c) accuracy

Direction Transform CMF → LF [Eq. 78]

$n_e = (\nu_0/\nu) \cdot [\, n_{0,e} + (1 + c\,\gamma / (c\,\gamma + 1)\, n_{0,e} \cdot v)\, \gamma\beta \,]$

- Applied after interaction in CMF
- Re-emission direction set in CMF first

Optical Depth Accumulation (Moving Media)

- Packets propagate in LF accumulating τ
- Line opacities transformed: $\chi_{LF} = \chi_{CMF} \cdot (\nu_0/\nu)$
- CMF frequency changes continuously along path
- General case requires integrating Eq. (79)

CMF Packet Initialisation

- Often natural to initialise packets in CMF
- e.g., thermal emission sampled from Planck in CMF
- Then transform to LF: apply Eqs. 75–78 in reverse
- Ensures correct energy conservation from the start

8.2 Line Interactions in Outflows: Sobolev Approximation

Simplifying bound-bound opacities in monotonically expanding flows

Bound-bound (line) processes are **resonant**: only photons with CMF frequency $\approx \nu_{lu}$ interact

In a monotonically expanding flow, photons redshift continuously — each line resonance is encountered **once**

- Over the line width, the plasma state is approximately constant → Sobolev approximation

Sobolev approximation: replace the full frequency-dependent line profile by a delta-function at ν_{lu}.
→ Optical depth integration collapses to a single local evaluation at the Sobolev point r_s.

Sobolev Optical Depth [Eq. 81]

$$\tau_s = (\pi e^2/m_e c) f_{lu} n_l (1 - n_u g_l / n_l g_u) (dl/d\nu_0)_{r_s}$$

- f_{lu}: absorption oscillator strength $l \to u$
- n_l, n_u: level populations
- g_l, g_u: statistical weights
- $(dl/d\nu_0)$: velocity gradient term [Eq. 82]

Homologous Flow Simplification [Eq. 82]

$$(dl/d\nu_0)_{r_s} = c/(\nu_{lu}) \cdot 1/((1-\mu^2) v/r + \mu^2 dv/dr)$$

- In homologous flow v/r = const
- Escape probability becomes direction-independent
- Greatly simplifies re-emission direction sampling

8.2 MCRT Implementation of the Sobolev Approximation

Packet propagation scheme with Sobolev line interactions

MC packets accumulate **continuum** optical depth τ_{cont} as they propagate (frequency-independent part)

Simultaneously, track distance to next **Sobolev point** for each line in a frequency-ordered list

- Decision at each step: does the packet reach a Sobolev point before τ_{cont} is exhausted?

Continuum Interaction (τ_{cont} reached first)

- Packet moved to distance l [Eq. 44]
- Interaction type sampled from continuum processes
- Sobolev line contributions reset/recalculated
- e.g., Thomson scattering, photoionization

Line Interaction (Sobolev point reached)

- Packet moved to Sobolev point r_s
- τ_{cont} instantaneously incremented by full τ_s
- Sample: scatter (resonant) or pass through
- Re-emission direction drawn from escape probability: $\rho(n) \propto (1 - \exp(-\tau_s(n))) / \tau_s(n)$ [Eq. 83]

The Sobolev approach stores lines in a frequency-ordered list (Abbott & Lucy 1985; Mazzali & Lucy 1993).
As the packet redshifts monotonically, it sweeps through lines one by one — a hugely efficient scheme.

8.2 Sobolev Optical Depth Accumulation

Fig. 7 — Illustration of the optical depth accumulation process

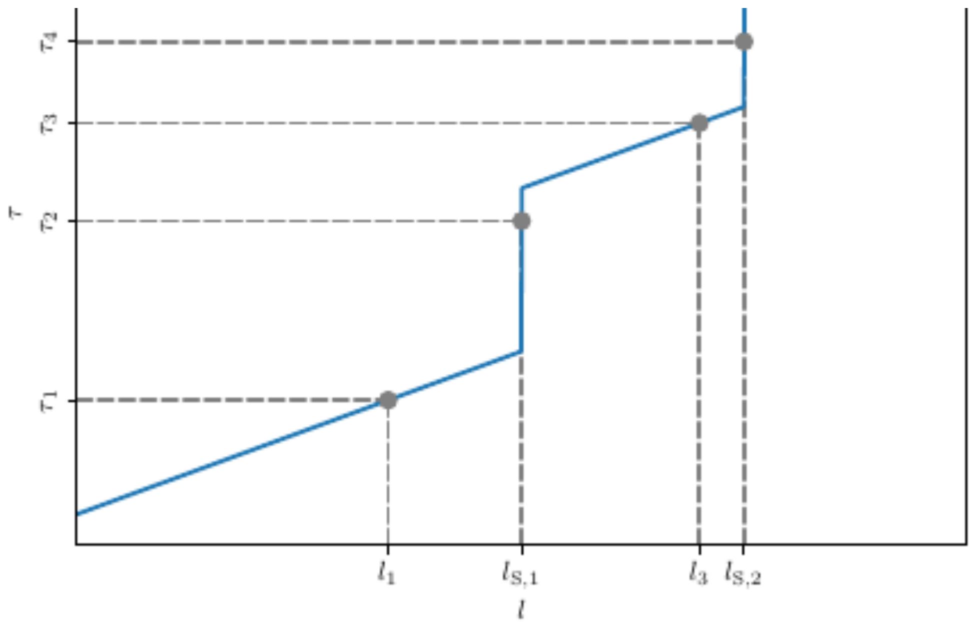

Fig. 7 Illustration of the optical depth accumulation process devised by Abbott and Lucy (1985) and Mazzali and Lucy (1993) for MCRT simulations relying on the Sobolev approximation. At the Sobolev points, S_1 and S_2, the accumulated optical depth is instantaneously incremented by the full line optical depth. In addition four possible outcomes of the decision process about the next interactions are shown. For τ_1 and τ_3, the packet would experience a continuum interaction at l_1 and l_3 respectively. In the remaining cases, i.e. for τ_2 and τ_4, the packet undergoes a line interaction at the respective Sobolev points $l_{S,1}$ and $l_{S,2}$. This figure is loosely adapted from Mazzali and Lucy (1993, Fig. 1)

Fig. 7: Optical depth τ accumulates continuously from continuum processes. At two Sobolev points S_1 and S_2, τ is instantaneously incremented by the full Sobolev line depth. For τ_1 and τ_3: packet undergoes a continuum interaction at l_1 or l_3. For τ_2 and τ_4: packet undergoes a line interaction at the corresponding Sobolev point.

will vary with direction and must be sampled whenever directions for re-emission of

8.2 Test: H Lyman α Line Profile in a Homologous Flow

Fig. 8 — Validation of the Sobolev-based MCRT scheme

U. M. Noebauer, S. A. Sim

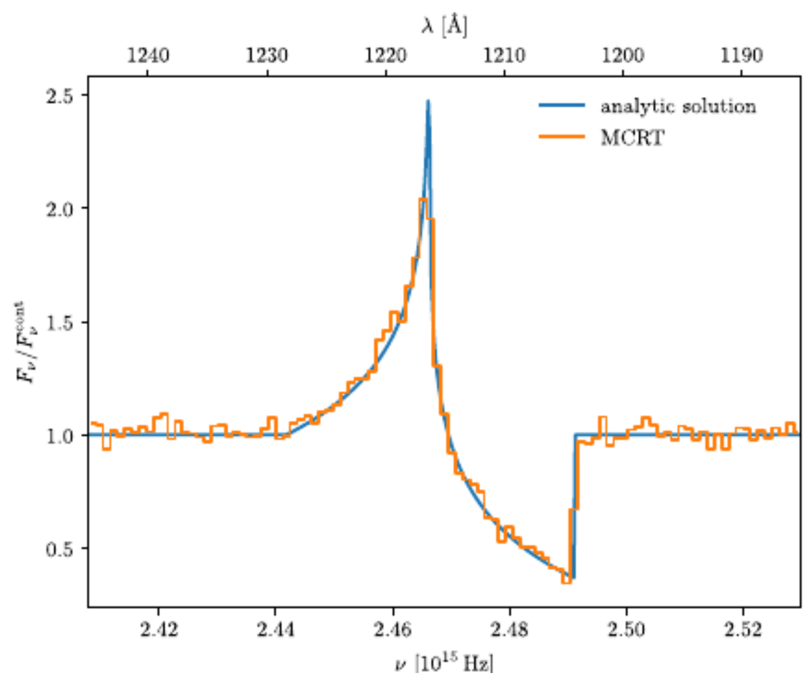

Fig. 8: H Lyman α line profile from a simple mixed-frame MCRT simulation. Blue: analytic solution (formal integration, Jeffery & Branch ... incid

Test Setup (Appendix A.2)

- H sphere, τ_s = 1 for Ly α, N = 10^5 packets
- Analytic: Jeffery & Branch (1990)

Result

- MCRT matches analytic solution closely
- MC noise visible but statistically valid

8.3 MCRT and Expansion Work

Energy exchange in mass outflows: the radiation field drives the expansion

In RE, packet energy is conserved in the **CMF** during interactions

But in the **LF**, energy is NOT conserved: packets can gain or lose energy at each scattering

Photons propagating **outward** in an outflow lose energy on average → this is **expansion work**

Mean LF energy after isotropic re-emission from a packet with incident direction μ_i (Eq. 87):

$\bar{\varepsilon}_e/\varepsilon_i = (1 - \mu_i \beta)\,[\log(1+\beta) - \log(1-\beta)] / (2\beta)$ → $\bar{\varepsilon}_e/\varepsilon_i \approx 1 - \mu_i \beta$ for small β

For $\mu_i = 1$ (outward propagating): $\bar{\varepsilon}_e < \varepsilon_i$ — net energy loss → expansion work!

Physical Interpretation

- Radiation field initially aligned with outflow
- After multiple scatterings, packets lose energy in LF
- Energy transferred to kinetic energy of the outflowing gas
- This is the radiative driving mechanism in hot-star winds

MCRT Naturally Captures This

- No special treatment needed in mixed-frame MCRT
- Expansion work = $\Sigma\,(\varepsilon_i - \varepsilon_e)$ over all interactions
- Tracked by comparing incident and emergent LF energies
- Lucy (2005): model SN test demonstrates this directly

8.3 Test: Lucy (2005) Model Supernova

Figs. 9 & 10 — Energy flows and expansion work in a simplified SN ejecta model

U. M. Noebauer, S. A. Sim

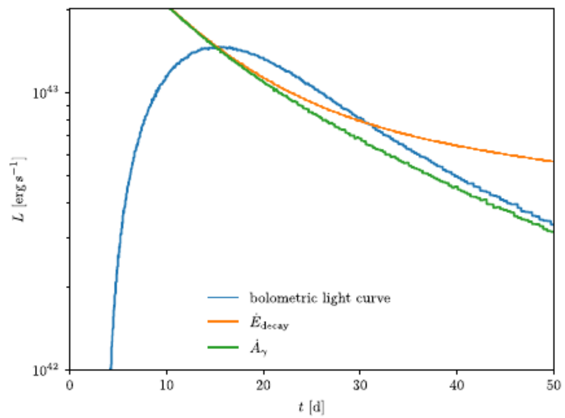

Fig. 9 Bolometric light curve obtained with the code MCRH (Noebauer et al. 2012) for the test problem devised by Lucy (2005). In addition to the evolution of the radiation emerging from the model SN (blue), the rate of energy released in the radioactive decay chain of ^{56}Ni (orange) and the rate at which γ packets deposit their energy in the ultraviolet-optical-infrared radiation field (green) are shown. See Appendix A.3

Monte Carlo radiative transfer

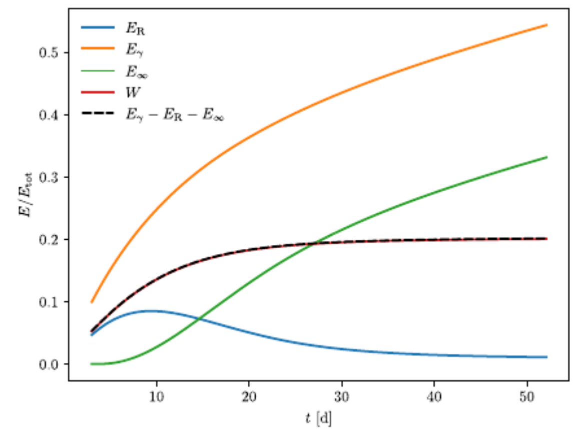

Fig. 10 Energy flows in the test problem devised by Lucy (2005). E_{γ} shows the energy that has been released by radioactive decays, E_{R} indicated the energy currently stored in the radiation field, and E_{∞} is the energy that has escaped to infinity. In addition to the work term, $W(t)$ obtained directly from the

Fig. 9 (left): Bolometric light curve from the MCRH code for the Lucy (2005) model SN. Blue: escaping luminosity. Orange: energy release rate from ^{56}Ni radioactive decay chain. Green: rate at which γ-packets deposit energy into the UV-optical-IR radiation field.

Fig. 10 (right): Energy flows — E_R(t) (radiation field energy), E_∞(t) (escaped energy), E_γ(t) (radioactive decay energy), W(t) (expansion work term). The black dashed curve E_γ − E_R − E_∞ precisely follows W, confirming energy conservation.

Key result: Mixed-frame MCRT correctly captures expansion work without special treatment.
Energy conservation verified at every time step across the simulation.

Chapter 8 — Summary

MCRT in outflows: mixed-frame, Sobolev, expansion work

Mixed-Frame Approach

- LF propagation + CMF interactions
- Doppler (Eqs. 71–76) + aberration (Eqs. 77–78)
- Optical depth accounts for CMF frequency drift
- Packet initialisation often in CMF

Sobolev Approximation

- Monotonic velocity field → each line once
- τ_s: purely local evaluation at Sobolev point
- Line interactions: sampled via escape probability
- H Lyman α test: excellent agreement

Expansion Work

- LF energy NOT conserved per interaction
- Net energy loss = expansion work done on gas
- MCRT captures this automatically
- Validated with Lucy (2005) model SN

- These techniques are widely used: Abbott & Lucy (1985), Mazzali & Lucy (1993), Vink et al. (1999, 2000), Sim (2004), Noebauer & Sim (2015)...

MCRT + Sobolev is **the** standard approach for spectral synthesis in SN ejecta and hot-star winds

Next: how to **extract** physical information from MCRT simulations more efficiently (Chapter 9)